

NAME

response -- Determine response calibrations

USAGE

response calibration normalization response

PARAMETERS**calibration**

Images to use in determining response calibrations. These are generally quartz continuum spectra. An image section may be used to select only a portion of the image.

normalization

Images to use determining the normalization spectrum. In almost all cases the normalization images are the same as the calibration images or a subsection of the calibration images.

responses

Response calibration images to be created. Each response image is with a calibration image. If the image exists then it will be modified otherwise it is created.

interactive= yes

Graph the average calibration spectrum and fit the normalization spectrum interactively?

threshold= INDEF

Set the response to 1 when the normalization spectrum or input image data fall below this value. If INDEF then no threshold is applied.

sample = ""

Sample of points to use in fitting the average calibration spectrum. The sample is selected with a range string.

naverage= 1

Number of sample points to average or median before fitting the function. If the number is positive the average of each set of naverage sample points is formed while if the number is negative then the median of each set of points (in absolute value) is formed. This subsample of points is used in fitting the normalization spectrum.

function = "spline3"

Function to fit to the average image spectrum to form the normalization spectrum. The options are "spline1", "spline3", "legendre", and "chebyshev".

order= 1

Order of the fitting function or the number of spline pieces.

low_reject= 0., high_reject= 0.

Rejection limits below and above the fit in units of the residual sigma.

niterate= 1

Number of rejection iterations.

grow = 0

Reject additional points within this distance of points exceeding the rejection threshold.

CURSOR KEYS

The interactive curve fitting package **icfit** is used to fit a to the average calibration spectrum. Help for this package is found under the name "icfit".

DESCRIPTION

A response calibration, in the form of an image, is created for each input image, normally a quartz spectrum. The response calibration is formed by dividing the calibration image by a normalization spectrum which is the same at all points along the spatial axis. The normalization spectrum is obtained by averaging the normalization image across the dispersion to form a one dimensional spectrum and smoothing the spectrum by fitting a function. The threshold value does not apply to creating or fitting of the normalization spectrum but only the final creation of the response values. When normalizing (that is dividing the data values by the fit to the normalization spectrum) only pixels in which both the fitted normalization value and the data value are above the threshold are computed. If either the normalization value or the data value is below the threshold the output response value is one.

The image header keyword DISPAXIS must be present with a dispersion parallel to the lines (varying with the column coordinate) or 2 for dispersion parallel to the columns (varying with line coordinate). This parameter may be added using **hedit**. Note that if the image has been transposed (**imtranspose**) the dispersion axis should still refer to the original dispersion axis unless the physical world coordinate system is first reset (see **wcsreset**). This is done in order to allow images which have DISPAXIS defined prior to transposing to still work correctly without requiring this keyword to be changed.

If the output image does not exist it is first created with unit response everywhere. Subsequently the response is only modified in those regions occupied by the input calibration image. Thus, image sections may be used to select regions in which the response is desired. This ability is particularly useful when dealing with multiple slits within an image or to exclude regions outside the slit.

Normally the normalization images are the same as the calibration images. In other words the calibration image is normalized by the average spectrum of the calibration image itself. Sometimes, however, the normalization image may be a smaller image section of the calibration image to avoid contaminating the normalization spectrum by effects at the edge of the slit. Again, this may be quite useful in multi-slit images.

The normalization spectrum is smoothed by fitting a using the interactive curve fitting package (**icfit**). The parameters determining the fitted normalization spectrum are the sample points, the averaging bin size, the fitting function, the order of the function, the rejection sigmas, the number of rejection iterations, and the rejection width. The sample points for the average spectrum are selected by a range string. Points in the normalization spectrum not in the sample are not used in determining the fitted function. The selected sample points may be binned into a set of averages or medians which are used in the function fit instead of the sample points with the averaging bin size parameter *naverage*. This parameter selects the number of sample points to be averaged if its value is positive or the number of points to be medianed if its value is negative (naturally, the absolute value is used for the number of points). A value of one uses all sample points without binning. The fitted function may be used to reject points from the fit using the parameters *low_reject*, *high_reject*, *niterate* and *grow*. If one or both of the rejection limits are greater than zero then the sigma of the residuals is computed and points with residuals less than *-low_reject* times the sigma and greater than *high_reject* times the sigma are removed and the function fitted again. In addition points within a distance given by the parameter *grow* of the a rejected point are also rejected. A value of zero for this parameter rejects only the points exceeding the rejection threshold. Finally, the rejection procedure may be iterated the number of times given by the parameter *niterate*.

The fitted function may be examined and modified interactively when the parameter *interactive* is set. In this case the normalization spectrum and the fitted function or the residuals of the fit are graphed. Deleted points are marked with an x and rejected points by a diamond. The sample regions are indicated along the bottom of the graph. The cursor keys and colon commands are used to change the values of the fitting

parameters, delete points, and window and expand the graph. When the fitted function is satisfactory exit with a carriage return or 'q' and the calibration image will be created. Changes in the fitted parameters are remembered from image to image within the task but not outside the task.

When the task finishes creating a response image the fitting parameters are updated in the parameter file.

EXAMPLES

1. To create a response image non-interactively:

```
cl> response quartz quartz response order=20 interactive=no
```

2. To determine independent responses for a image sections defining each slit. Then the responses are computed as follows:

```
cl> response quartz[10:20,*],quartz[35:45,*] \  
>>> quartz[12:18,*],quartz[12:18,*] resp,resp
```

Generally the slit image sections are prepared in a used to define the lists of input images and response.

```
cl> response @slits @slits @responses
```

3. If the DISPAXIS keyword is missing and the dispersion is running vertically (varying with the image lines):

```
cl> hedit *.imh dispaxis 2 add+
```

SEE ALSO

icfit, iillumination